DBMS LEVEL 3 PROJECT SYNOPSIS

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Project Title:

Smart Research & Peer Review Portal

1. Problem Statement

Traditional academic research and peer-review systems mainly depend on relational databases and keyword-based searches. Keyword searches may fail to identify research papers with similar meanings but different terminology. Manual reviewer assignment can also be time-consuming.

The proposed **Smart Research & Peer Review Portal** combines PostgreSQL with vector-based semantic search to efficiently manage research manuscripts, reviewers, reviews, and academic content.

2. Objectives

- To develop a web-based research and peer-review portal using **Next.js**.
- To use **PostgreSQL** for storing users, manuscripts, reviewers, reviews, scores, and submission status.
- To use **pgvector** for storing manuscript embeddings and performing semantic similarity searches.
- To implement **Student, Reviewer, and Admin** roles with role-based access.

- To provide manuscript submission, reviewer assignment, review evaluation, and status tracking.
- To demonstrate SQL operations such as **INSERT, UPDATE, DELETE, SELECT, JOIN, GROUP BY, HAVING, ORDER BY, aggregate functions, and subqueries**.
- To implement advanced database concepts such as **transactions, triggers, functions/procedures, views, indexes, and vector similarity search**.

3. Proposed System

The portal will provide three main user roles:

Student

- Register and login.
- Submit research manuscripts.
- Track manuscript status.
- View permitted review results.

Reviewer

- Login securely.
- View assigned manuscripts.
- Submit review comments and scores.

Admin

- Manage users and reviewers.
- View submitted manuscripts.
- Assign reviewers.
- Monitor reviews and manuscript status.
- View review analytics.
- Use semantic similarity results to assist reviewer selection.

4. Database Architecture

The project will use a hybrid database approach:

PostgreSQL:
Stores structured information such as users, roles, manuscripts, reviewers, assignments, reviews, scores, and audit records.

pgvector:
Stores vector embeddings generated from manuscript abstracts and performs similarity searches to identify related research and assist with reviewer matching.

The SQL database will use **primary keys, foreign keys, constraints, indexes, and transactions** to maintain data integrity. An ER diagram and relational schema will also be prepared.

5. Semantic Search

When a manuscript is submitted, its abstract will be converted into a vector embedding and stored using **pgvector**.

The system will use **cosine similarity** to identify semantically related manuscripts. Similarity results can also help administrators identify reviewers whose research expertise is related to a submitted manuscript.

The project will document the embedding approach, vector storage, similarity metric, and retrieval process as required.

6. Advanced Database Features

The project will demonstrate:

1. **Transactions** – for safe submission and review workflows.
2. **Triggers** – for automated database actions and audit records.
3. **Functions/Procedures** – for reusable database operations.
4. **Views/Materialized Views** – for reports and review analytics.
5. **Indexes** – for improving query performance.
6. **Vector Similarity Search** – for semantic research discovery.

This exceeds the requirement of demonstrating at least four advanced database concepts.

7. Technology Stack

- **Frontend:** Next.js / React
- **Backend/API:** Next.js Route Handlers
- **SQL Database:** PostgreSQL
- **Vector Database:** PostgreSQL with pgvector
- **Version Control:** Git & GitHub

The project requirements specifically support PostgreSQL with pgvector and require a functional Next.js web interface.

8. Security

The system will implement:

- Secure authentication.
- Role-based authorization.
- Password hashing.
- Input validation.
- Database constraints.
- Environment variables for database credentials.
- Protection against unauthorized access.

Passwords and database credentials will not be stored in plain text.

9. Expected Outcome

The final **Smart Research & Peer Review Portal** will provide a functional platform for managing academic research submissions and peer review. It will combine reliable PostgreSQL database management with semantic vector search to improve research discovery and assist reviewer assignment.

The project will demonstrate practical **database design, SQL operations, advanced DBMS concepts, vector similarity search, authentication, and Next.js application development**, going beyond basic CRUD functionality as required for the Level 3 project.